

TOC-2026: Quiz-4 (Decision Properties – DFA)

Full Marks: 15, **Time:** 40 minutes

(1) Let M be a DFA over a finite alphabet Σ and consider the following language

$$\text{ALL}_{\text{DFA}} = \{\langle M \rangle \mid M \text{ is a DFA over } \Sigma \text{ and } L(M) = \Sigma^*\}.$$

Design an algorithm that correctly decides ALL_{DFA} .

(8 Marks)

(2) Let B be the set of encodings of all NFAs that accept at least one string from $\{1\}^*$. Formally, define $B = \{\langle M \rangle \mid L(M) \cap \{1\}^* \neq \emptyset\}$.

Design an algorithm that correctly decides B .

(7 Marks)